

1. Tumoral dynamics

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1 Environment preparation

Run scripts with the needed packages, functions, and the raw data frame.

```
source("0a_packages.R")
source("0b_script_data.R")
source("0c_functions.R")
```

2 Data formatting

Create a new data frame adapted for survival analysis.

```
tum_df <-

#' Load the raw data frame.
df %>%

#' Restrict the data frame to the three lineages that developed tumors.
filter(lineage %in% c("HO_MT", "HO_SPC", "HO_SPT")) %>%

#' Define the follow-up time for each individual:
#' if the hydra developed a tumor -> time of tumor onset (event),
#' if the hydra died -> time of death (sensor),
#' if the hydra did not develop a tumor and stayed alive
#' -> 42 days (no event, sensor at the end of the experiment).
mutate(age_followup_days = case_when(
  tum_state == "Tumoral" ~ age_tum,
  tum_state == "Healthy" & death_state == 1 ~ age_death,
  TRUE ~ 42),

age_followup = age_followup_days / 7,
tum_event = ifelse(tum_state == "Tumoral", 1, 0)) %>%
```

```

#' Build a competing-risk variable for the cumulative incidence functions:
#' if the hydra did not develop a tumor and stayed alive -> 0 (no event),
#' if the hydra developed a tumor -> 1 (event),
#' if the hydra died -> 2 (competing risk of death).
mutate(status_competing = factor(case_when(tum_event == 1 ~ 1,
                                           death_state == 1 ~ 2,
                                           TRUE ~ 0))) %>%

#' Remove missing data.
filter(!is.na(age_followup)) %>%

#' Set factor levels for later use in the plots.
mutate(treatment_condition = factor(treatment_condition,
                                    levels = c("Control", "Shaking treatment"))) %>%

#' Keep only the variables needed for the analysis.
select(age_followup, tum_event, status_competing, treatment_condition,
       lineage, parent_unique, replicate_unique)

```

Set reference levels for the categorical variables.

```

tum_df$lineage <- relevel(factor(tum_df$lineage), ref = "HO_MT")
tum_df$treatment_condition <- relevel(factor(tum_df$treatment_condition), ref =
"Control")

```

3 Survival object

Create a survival object with follow-up time and event indicator.

```
tum_surv_object <- Surv(tum_df$age_followup, tum_df$tum_event)
```

4 Proportional hazards assumption check : Schoenfeld residuals

Fit a preliminary full Cox model (fixed effects only).

```
tum_cox <- coxph(Surv(age_followup, tum_event) ~ treatment_condition * lineage, data =
tum_df)
```

Test the proportional hazards assumption.

```
print(cox.zph(tum_cox))
```

```
##                chisq df      p
## treatment_condition    0.575  1 0.448
## lineage                5.279  2 0.071
## treatment_condition:lineage 8.986  2 0.011
## GLOBAL                 11.395  5 0.044
```

The proportional hazards assumption is violated for at least one term, an accelerated failure time mixed-effects model will be used.

5 AIC comparison of different distributions

Compare intercept-only accelerated failure time models fitted with different parametric distributions to identify the best-fitting one.

```
dist_weib <- SurvregME(Surv(age_followup, tum_event) ~ 1, data = tum_df,
  dist = "weibull")

dist_lnorm <- SurvregME(Surv(age_followup, tum_event) ~ 1, data = tum_df,
  dist = "lognormal")

dist_llog <- SurvregME(Surv(age_followup, tum_event) ~ 1, data = tum_df,
  dist = "loglogistic")

dist_exp <- SurvregME(Surv(age_followup, tum_event) ~ 1, data = tum_df,
  dist = "exponential")

AIC(dist_weib, dist_lnorm, dist_llog, dist_exp) %>% arrange(AIC)
```

```
##      df      AIC
## dist_weib  2 1291.277
## dist_llog  2 1294.009
## dist_lnorm 2 1334.860
## dist_exp   1 1347.392
```

The model with a Weibull distribution has the lowest AIC, we'll use it for the models.

6 Model selection

6.1 Random effects selection

With the full fixed-effect structure held constant, compare different random-effect structures to find the optimal one.

```
tum_R_0 <- SurvregME(Surv(age_followup, tum_event) ~ treatment_condition * lineage,
  data = tum_df, dist = "weibull")

tum_R_1 <- SurvregME(Surv(age_followup, tum_event) ~ treatment_condition * lineage +
  (1 | parent_unique),
  data = tum_df, dist = "weibull")

tum_R_2 <- SurvregME(Surv(age_followup, tum_event) ~ treatment_condition * lineage +
  (1 | replicate_unique),
  data = tum_df, dist = "weibull")

tum_R_3 <- SurvregME(Surv(age_followup, tum_event) ~ treatment_condition * lineage +
  (1 | parent_unique) + (1 | replicate_unique),
```

```
data = tum_df, dist = "weibull")

AIC(tum_R_0, tum_R_1, tum_R_2, tum_R_3) %>% arrange(AIC)
```

```
##      df      AIC
## tum_R_0  7 1157.217
## tum_R_2  8 1158.027
## tum_R_1  8 1159.087
## tum_R_3  9 1160.027
```

The model without any random effect (tum_R_0) has the lowest AIC, this random effect structure is therefore selected. SurvregME is nonetheless kept as the modeling function (rather than switching to a simpler survreg call) since it also supports models with no random effects.

6.2 Fixed effects selection

With the selected random-effect structure fixed, compare different fixed-effect structures to find the optimal one.

```
tum_F_0 <- SurvregME(Surv(age_followup, tum_event) ~ 1,
                     data = tum_df, dist = "weibull")

tum_F_1 <- SurvregME(Surv(age_followup, tum_event) ~ treatment_condition,
                     data = tum_df, dist = "weibull")

tum_F_2 <- SurvregME(Surv(age_followup, tum_event) ~ lineage,
                     data = tum_df, dist = "weibull")

tum_F_3 <- SurvregME(Surv(age_followup, tum_event) ~ treatment_condition + lineage,
                     data = tum_df, dist = "weibull")

tum_F_4 <- SurvregME(Surv(age_followup, tum_event) ~ treatment_condition * lineage,
                     data = tum_df, dist = "weibull")

AIC(tum_F_0, tum_F_1, tum_F_2, tum_F_3, tum_F_4) %>% arrange(AIC)
```

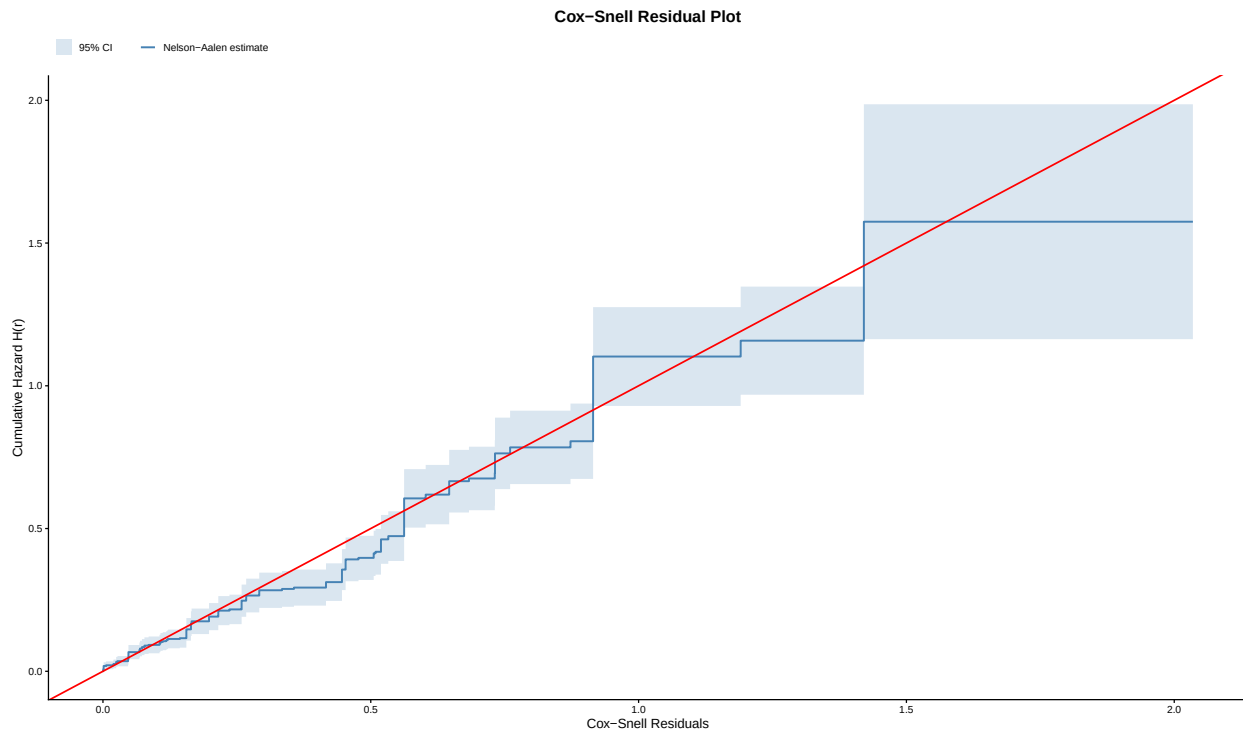
```
##      df      AIC
## tum_F_4  7 1157.217
## tum_F_3  5 1203.737
## tum_F_2  4 1212.049
## tum_F_1  3 1275.994
## tum_F_0  2 1291.277
```

The model with the interaction between the treatment condition and the lineage has the lowest AIC and is therefore selected (tum_F_4).

6.3 Goodness-of-fit evaluation

Assess goodness-of-fit of the selected model using Cox-Snell residuals.

```
plot_cox_snell_survregme(model = tum_F_4, data = tum_df,
                        time_col = "age_followup", event_col = "tum_event")
```



The curve follows a 45 degree exponential, we validate the fit of the selected model.

6.4 Selected model summary

Releveling of “lineage” to obtain all pairwise comparisons from the same selected model.

```
tum_df$lineage <- relevel(tum_df$lineage, ref = "HO_MT")
summary(SurvregME(Surv(age_followup, tum_event) ~ treatment_condition * lineage,
                  data = tum_df, dist = "weibull"))
```

```
##
## Weibull Linear Regression Model
##
## Formula:
## Surv(age_followup, tum_event) ~ treatment_condition * lineage
##
## Fitted to dataset tum_df
##
## Fixed effects parameters:
## =====
##
##
```

	Estimate	Std. Error	z value
treatment_conditionShaking treatment	-1.439789	0.359848	-4.0011
lineageHO_SPC	0.016721	0.447214	0.0374
lineageHO_SPT	-2.582680	0.343763	-7.5130

```
## treatment_conditionShaking treatment:lineageHO_SPC -0.372221 0.502700 -0.7404
## treatment_conditionShaking treatment:lineageHO_SPT 1.926839 0.409974 4.6999
## Pr(>|z|)
## treatment_conditionShaking treatment 6.305e-05 ***
## lineageHO_SPC 0.9702
## lineageHO_SPT 5.780e-14 ***
## treatment_conditionShaking treatment:lineageHO_SPC 0.4590
## treatment_conditionShaking treatment:lineageHO_SPT 2.603e-06 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Log-likelihood: -571.6086 (npar = 7)
```

```
tum_df$lineage <- relevel(tum_df$lineage, ref = "HO_SPC")
summary(SurvregME(Surv(age_followup, tum_event) ~ treatment_condition * lineage,
  data = tum_df, dist = "weibull"))
```

```
##
## Weibull Linear Regression Model
##
## Formula:
## Surv(age_followup, tum_event) ~ treatment_condition * lineage
##
## Fitted to dataset tum_df
##
## Fixed effects parameters:
## =====
##
## Estimate Std. Error z value
## treatment_conditionShaking treatment -1.812001 0.351408 -5.1564
## lineageHO_MT -0.016705 0.447214 -0.0374
## lineageHO_SPT -2.599394 0.343686 -7.5633
## treatment_conditionShaking treatment:lineageHO_MT 0.372206 0.502700 0.7404
## treatment_conditionShaking treatment:lineageHO_SPT 2.299055 0.402866 5.7067
## Pr(>|z|)
## treatment_conditionShaking treatment 2.517e-07 ***
## lineageHO_MT 0.9702
## lineageHO_SPT 3.930e-14 ***
## treatment_conditionShaking treatment:lineageHO_MT 0.4590
## treatment_conditionShaking treatment:lineageHO_SPT 1.152e-08 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Log-likelihood: -571.6086 (npar = 7)
```

```
tum_df$lineage <- relevel(tum_df$lineage, ref = "HO_SPT")
summary(SurvregME(Surv(age_followup, tum_event) ~ treatment_condition * lineage,
  data = tum_df, dist = "weibull"))
```

```
##
## Weibull Linear Regression Model
##
```

```
## Formula:
## Surv(age_followup, tum_event) ~ treatment_condition * lineage
##
## Fitted to dataset tum_df
##
## Fixed effects parameters:
## =====
##
##                                     Estimate Std. Error z value
## treatment_conditionShaking treatment      0.48705    0.19562   2.4898
## lineageHO_SPC                          2.59940    0.34369   7.5633
## lineageHO_MT                           2.58269    0.34376   7.5130
## treatment_conditionShaking treatment:lineageHO_SPC -2.29906    0.40287  -5.7068
## treatment_conditionShaking treatment:lineageHO_MT  -1.92685    0.40997  -4.6999
##                                     Pr(>|z|)
## treatment_conditionShaking treatment      0.01278 *
## lineageHO_SPC                          3.930e-14 ***
## lineageHO_MT                           5.780e-14 ***
## treatment_conditionShaking treatment:lineageHO_SPC 1.152e-08 ***
## treatment_conditionShaking treatment:lineageHO_MT  2.603e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Log-likelihood: -571.6086 (npar = 7)
```

7 Median times

7.1 By treatment condition

Estimate cumulative incidence functions for the tumor event, accounting for the competing risk of death, stratified by treatment condition.

```
tum_cuminc_treatment <- cuminc(Surv(age_followup, status_competing) ~
                               treatment_condition, data = tum_df)
```

For each stratum, extract the median time to event and its 95CI.

```
tum_medians_treatment <- tidy(tum_cuminc_treatment) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::group_by(strata) %>%
  dplyr::arrange(time) %>%
  dplyr::summarise(
    median_time = { idx <- which(estimate >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_low = { idx <- which(conf.high >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_high = { idx <- which(conf.low >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    .groups = "drop")

print(tum_medians_treatment)
```



```
## # A tibble: 2 x 4
##   strata      median_time ci_low ci_high
##   <fct>          <dbl>   <dbl>   <dbl>
## 1 Control              NA     NA       NA
## 2 Shaking treatment      5    4.14    5.43
```

7.2 By lineage

Same method as above, but stratified by lineage.

```
tum_cuminc_lineage <- cuminc(Surv(age_followup, status_competing) ~ lineage, data =
tum_df)

tum_medians_lineage <- tidy(tum_cuminc_lineage) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::group_by(strata) %>%
  dplyr::arrange(time) %>%
  dplyr::summarise(
    median_time = { idx <- which(estimate >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_low = { idx <- which(conf.high >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_high = { idx <- which(conf.low >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    .groups = "drop")

print(tum_medians_lineage)
```

```
## # A tibble: 3 x 4
##   strata median_time ci_low ci_high
##   <fct>          <dbl>   <dbl>   <dbl>
## 1 HO_SPT          4      4      4
## 2 HO_SPC          NA     NA     NA
## 3 HO_MT           NA     NA     NA
```

7.3 By treatment condition and lineage

Same method as above, but stratified by treatment condition and lineage.

```
tum_cuminc_interaction <- cuminc(Surv(age_followup, status_competing) ~
  treatment_condition + lineage, data = tum_df)

tum_medians_interaction <- tidy(tum_cuminc_interaction) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::group_by(strata) %>%
  dplyr::arrange(time) %>%
  dplyr::summarise(
    median_time = { idx <- which(estimate >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_low = { idx <- which(conf.high >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_high = { idx <- which(conf.low >= 0.5)[1]
```

```

    if (is.na(idx)) NA_real_ else time[idx] },
    .groups = "drop")

print(tum_medians_interaction)

## # A tibble: 6 x 4
##   strata                median_time ci_low ci_high
##   <fct>                <dbl>   <dbl>   <dbl>
## 1 Control, HO_SPT              4     4       4
## 2 Shaking treatment, HO_SPT    4     4     4.57
## 3 Control, HO_SPC             NA    NA      NA
## 4 Shaking treatment, HO_SPC     5   4.14    NA
## 5 Control, HO_MT              NA    NA      NA
## 6 Shaking treatment, HO_MT     NA     5      NA

```

8 Cumulative incidences

8.1 By week and treatment condition

Convert the continuous cumulative incidence function into a weekly summary table: for each stratum of treatment condition and each week, show the estimate with its 95CI.

```

tum_incidence_treatment <- tidy(tum_cuminc_treatment) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::mutate(week = ceiling(time)) %>%
  dplyr::group_by(strata, week) %>%
  dplyr::filter(time == max(time)) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
  dplyr::select(strata, week, estimate_cuminc = estimate, conf.low, conf.high) %>%
  dplyr::arrange(strata, week)

print(tum_incidence_treatment)

```

```

## # A tibble: 14 x 5
##   strata                week estimate_cuminc conf.low conf.high
##   <fct>                <dbl>         <dbl>     <dbl>     <dbl>
## 1 Control              0           0      NA      NA
## 2 Control              1       0.0278  0.0115  0.0565
## 3 Control              2       0.0417  0.0205  0.0745
## 4 Control              3       0.102   0.0660  0.147
## 5 Control              4       0.296   0.237   0.358
## 6 Control              5       0.356   0.293   0.420
## 7 Control              6       0.370   0.306   0.435
## 8 Shaking treatment    0           0      NA      NA
## 9 Shaking treatment    1       0.0185  0.00615 0.0439
## 10 Shaking treatment   2       0.0833  0.0513  0.125
## 11 Shaking treatment   3       0.167   0.120   0.219
## 12 Shaking treatment   4       0.384   0.319   0.449
## 13 Shaking treatment   5       0.537   0.468   0.601
## 14 Shaking treatment   6       0.574   0.505   0.637

```

8.2 By week and lineage

Same method as above, but stratified by lineage.

```
tum_incidence_lineage <- tidy(tum_cuminc_lineage) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::mutate(week = ceiling(time)) %>%
  dplyr::group_by(strata, week) %>%
  dplyr::filter(time == max(time)) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
  dplyr::select(strata, week, estimate_cuminc = estimate, conf.low, conf.high) %>%
  dplyr::arrange(strata, week)

print(tum_incidence_lineage)
```

```
## # A tibble: 21 x 5
##   strata week estimate_cuminc conf.low conf.high
##   <fct> <dbl>         <dbl>     <dbl>     <dbl>
## 1 HO_SPT  0           0         NA         NA
## 2 HO_SPT  1       0.0486    0.0214    0.0926
## 3 HO_SPT  2       0.139     0.0883    0.201
## 4 HO_SPT  3       0.257     0.189     0.331
## 5 HO_SPT  4       0.653     0.569     0.725
## 6 HO_SPT  5       0.743     0.663     0.807
## 7 HO_SPT  6       0.743     0.663     0.807
## 8 HO_SPC  0           0         NA         NA
## 9 HO_SPC  1     0.00694  0.000628  0.0351
##10 HO_SPC  2     0.00694  0.000628  0.0351
##11 HO_SPC  3     0.0833   0.0454    0.136
##12 HO_SPC  4     0.181    0.123     0.248
##13 HO_SPC  5     0.333    0.257     0.411
##14 HO_SPC  6     0.368    0.290     0.447
##15 HO_MT  0           0         NA         NA
##16 HO_MT  1     0.0139   0.00272   0.0452
##17 HO_MT  2     0.0417   0.0171   0.0836
##18 HO_MT  3     0.0625   0.0306   0.110
##19 HO_MT  4     0.188    0.128     0.255
##20 HO_MT  5     0.264    0.195     0.338
##21 HO_MT  6     0.306    0.232     0.382
```

8.3 By week, treatment condition, and lineage

Same method as above, but stratified by treatment condition and lineage.

```
tum_incidence_interaction <- tidy(tum_cuminc_interaction) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::mutate(week = ceiling(time)) %>%
  dplyr::group_by(strata, week) %>%
  dplyr::filter(time == max(time)) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
```

```
dplyr::select(strata, week, estimate_cuminc = estimate, conf.low, conf.high) %>%
dplyr::arrange(strata, week)

print(tum_incidence_interaction)
```

```
## # A tibble: 42 x 5
##   strata                week estimate_cuminc conf.low conf.high
##   <fct>                <dbl>         <dbl>    <dbl>    <dbl>
## 1 Control, HO_SPT      0           0      NA      NA
## 2 Control, HO_SPT      1       0.0694  0.0255  0.144
## 3 Control, HO_SPT      2       0.0972  0.0425  0.179
## 4 Control, HO_SPT      3       0.264   0.168   0.370
## 5 Control, HO_SPT      4       0.736   0.617   0.824
## 6 Control, HO_SPT      5       0.833   0.722   0.903
## 7 Control, HO_SPT      6       0.833   0.722   0.903
## 8 Shaking treatment, HO_SPT 0           0      NA      NA
## 9 Shaking treatment, HO_SPT 1       0.0278  0.00519 0.0871
## 10 Shaking treatment, HO_SPT 2       0.181   0.102   0.278
## 11 Shaking treatment, HO_SPT 3       0.25    0.157   0.355
## 12 Shaking treatment, HO_SPT 4       0.569   0.446   0.675
## 13 Shaking treatment, HO_SPT 5       0.653   0.529   0.751
## 14 Shaking treatment, HO_SPT 6       0.653   0.529   0.751
## 15 Control, HO_SPC      0           0      NA      NA
## 16 Control, HO_SPC      1           0      NA      NA
## 17 Control, HO_SPC      2           0      NA      NA
## 18 Control, HO_SPC      3       0.0139  0.00116 0.0669
## 19 Control, HO_SPC      4       0.0556  0.0178  0.126
## 20 Control, HO_SPC      5       0.111   0.0516  0.196
## 21 Control, HO_SPC      6       0.139   0.0708  0.229
## 22 Shaking treatment, HO_SPC 0           0      NA      NA
## 23 Shaking treatment, HO_SPC 1       0.0139  0.00116 0.0669
## 24 Shaking treatment, HO_SPC 2       0.0139  0.00116 0.0669
## 25 Shaking treatment, HO_SPC 3       0.153   0.0809  0.246
## 26 Shaking treatment, HO_SPC 4       0.306   0.203   0.414
## 27 Shaking treatment, HO_SPC 5       0.556   0.432   0.662
## 28 Shaking treatment, HO_SPC 6       0.597   0.473   0.701
## 29 Control, HO_MT       0           0      NA      NA
## 30 Control, HO_MT       1       0.0139  0.00116 0.0669
## 31 Control, HO_MT       2       0.0278  0.00519 0.0871
## 32 Control, HO_MT       3       0.0278  0.00519 0.0871
## 33 Control, HO_MT       4       0.0972  0.0424  0.179
## 34 Control, HO_MT       5       0.125   0.0611  0.213
## 35 Control, HO_MT       6       0.139   0.0708  0.229
## 36 Shaking treatment, HO_MT 0           0      NA      NA
## 37 Shaking treatment, HO_MT 1       0.0139  0.00116 0.0669
## 38 Shaking treatment, HO_MT 2       0.0556  0.0178  0.126
## 39 Shaking treatment, HO_MT 3       0.0972  0.0424  0.179
## 40 Shaking treatment, HO_MT 4       0.278   0.180   0.385
## 41 Shaking treatment, HO_MT 5       0.403   0.289   0.514
## 42 Shaking treatment, HO_MT 6       0.472   0.353   0.583
```

9 Cumulative proportions

9.1 By week and treatment condition

Compute raw cumulative proportions of tumor occurrence per week, stratified by treatment condition.

```
tum_proportions_treatment <- tum_df %>%
  dplyr::mutate(week_tum_event = ceiling(age_followup)) %>%
  dplyr::group_by(treatment_condition) %>%
  dplyr::summarise(total_N = n(),
    n1 = sum(week_tum_event <= 1 & tum_event == 1),
    n2 = sum(week_tum_event <= 2 & tum_event == 1),
    n3 = sum(week_tum_event <= 3 & tum_event == 1),
    n4 = sum(week_tum_event <= 4 & tum_event == 1),
    n5 = sum(week_tum_event <= 5 & tum_event == 1),
    n6 = sum(week_tum_event <= 6 & tum_event == 1),
    p1 = n1 / total_N,
    p2 = n2 / total_N,
    p3 = n3 / total_N,
    p4 = n4 / total_N,
    p5 = n5 / total_N,
    p6 = n6 / total_N,
    .groups = "drop") %>%
  tidyr::pivot_longer(cols = matches("[np][1-6]$"),
    names_to = c(".value", "week"),
    names_pattern = "(n|p)([1-6])") %>%
  dplyr::mutate(week = as.numeric(week)) %>%
  dplyr::select(treatment_condition, week, n, total_N, raw_proportion = p)

print(tum_proportions_treatment)
```

```
## # A tibble: 12 x 5
##   treatment_condition week      n total_N raw_proportion
##   <fct>              <dbl> <int>   <int>         <dbl>
## 1 Control            1      6    216         0.0278
## 2 Control            2      9    216         0.0417
## 3 Control            3     22    216         0.102
## 4 Control            4     64    216         0.296
## 5 Control            5     77    216         0.356
## 6 Control            6     80    216         0.370
## 7 Shaking treatment  1      4    216         0.0185
## 8 Shaking treatment  2     18    216         0.0833
## 9 Shaking treatment  3     36    216         0.167
## 10 Shaking treatment 4     83    216         0.384
## 11 Shaking treatment 5    116    216         0.537
## 12 Shaking treatment 6    124    216         0.574
```

9.2 By week and lineage

Same method as above, but stratified by lineage.

```

tum_proportions_lineage <- tum_df %>%
  dplyr::mutate(week_tum_event = ceiling(age_followup)) %>%
  dplyr::group_by(lineage) %>%
  dplyr::summarise(total_N = n(),
    n1 = sum(week_tum_event <= 1 & tum_event == 1),
    n2 = sum(week_tum_event <= 2 & tum_event == 1),
    n3 = sum(week_tum_event <= 3 & tum_event == 1),
    n4 = sum(week_tum_event <= 4 & tum_event == 1),
    n5 = sum(week_tum_event <= 5 & tum_event == 1),
    n6 = sum(week_tum_event <= 6 & tum_event == 1),
    p1 = n1 / total_N,
    p2 = n2 / total_N,
    p3 = n3 / total_N,
    p4 = n4 / total_N,
    p5 = n5 / total_N,
    p6 = n6 / total_N,
    .groups = "drop") %>%
  tidyr::pivot_longer(cols = matches("^np[1-6]$"),
    names_to = c(".value", "week"),
    names_pattern = "(n|p)([1-6])") %>%
  dplyr::mutate(week = as.numeric(week)) %>%
  dplyr::select(lineage, week, n, total_N, raw_proportion = p)

print(tum_proportions_lineage)

```

```

## # A tibble: 18 x 5
##   lineage week      n total_N raw_proportion
##   <fct>   <dbl> <int>   <int>         <dbl>
## 1 HO_SPT     1      7    144         0.0486
## 2 HO_SPT     2     20    144         0.139
## 3 HO_SPT     3     37    144         0.257
## 4 HO_SPT     4     94    144         0.653
## 5 HO_SPT     5    107    144         0.743
## 6 HO_SPT     6    107    144         0.743
## 7 HO_SPC     1      1    144         0.00694
## 8 HO_SPC     2      1    144         0.00694
## 9 HO_SPC     3     12    144         0.0833
## 10 HO_SPC    4     26    144         0.181
## 11 HO_SPC    5     48    144         0.333
## 12 HO_SPC    6     53    144         0.368
## 13 HO_MT     1      2    144         0.0139
## 14 HO_MT     2      6    144         0.0417
## 15 HO_MT     3      9    144         0.0625
## 16 HO_MT     4     27    144         0.188
## 17 HO_MT     5     38    144         0.264
## 18 HO_MT     6     44    144         0.306

```

9.3 By week, treatment condition, and lineage

Same method as above, but stratified by treatment condition and lineage.

```

tum_proportions_interaction <- tum_df %>%
  dplyr::mutate(week_tum_event = ceiling(age_followup)) %>%
  dplyr::group_by(treatment_condition, lineage) %>%
  dplyr::summarise(total_N = n(),
    n1 = sum(week_tum_event <= 1 & tum_event == 1),
    n2 = sum(week_tum_event <= 2 & tum_event == 1),
    n3 = sum(week_tum_event <= 3 & tum_event == 1),
    n4 = sum(week_tum_event <= 4 & tum_event == 1),
    n5 = sum(week_tum_event <= 5 & tum_event == 1),
    n6 = sum(week_tum_event <= 6 & tum_event == 1),
    p1 = n1 / total_N,
    p2 = n2 / total_N,
    p3 = n3 / total_N,
    p4 = n4 / total_N,
    p5 = n5 / total_N,
    p6 = n6 / total_N,
    .groups = "drop") %>%
  tidyr::pivot_longer(cols = matches("^np[1-6]$"),
    names_to = c(".value", "week"),
    names_pattern = "(n|p)([1-6])" %>%
  dplyr::mutate(week = as.numeric(week)) %>%
  dplyr::select(treatment_condition, lineage, week, n, total_N, raw_proportion = p)

print(tum_proportions_interaction)

```

```

## # A tibble: 36 x 6
##   treatment_condition lineage  week      n total_N raw_proportion
##   <fct>                <fct>  <dbl> <int>   <int>      <dbl>
## 1 Control              HO_SPT    1      5     72      0.0694
## 2 Control              HO_SPT    2      7     72      0.0972
## 3 Control              HO_SPT    3     19     72      0.264
## 4 Control              HO_SPT    4     53     72      0.736
## 5 Control              HO_SPT    5     60     72      0.833
## 6 Control              HO_SPT    6     60     72      0.833
## 7 Control              HO_SPC    1      0     72      0
## 8 Control              HO_SPC    2      0     72      0
## 9 Control              HO_SPC    3      1     72      0.0139
## 10 Control             HO_SPC    4      4     72      0.0556
## 11 Control             HO_SPC    5      8     72      0.111
## 12 Control             HO_SPC    6     10     72      0.139
## 13 Control             HO_MT     1      1     72      0.0139
## 14 Control             HO_MT     2      2     72      0.0278
## 15 Control             HO_MT     3      2     72      0.0278
## 16 Control             HO_MT     4      7     72      0.0972
## 17 Control             HO_MT     5      9     72      0.125
## 18 Control             HO_MT     6     10     72      0.139
## 19 Shaking treatment   HO_SPT    1      2     72      0.0278
## 20 Shaking treatment   HO_SPT    2     13     72      0.181
## 21 Shaking treatment   HO_SPT    3     18     72      0.25
## 22 Shaking treatment   HO_SPT    4     41     72      0.569
## 23 Shaking treatment   HO_SPT    5     47     72      0.653
## 24 Shaking treatment   HO_SPT    6     47     72      0.653
## 25 Shaking treatment   HO_SPC    1      1     72      0.0139
## 26 Shaking treatment   HO_SPC    2      1     72      0.0139

```

## 27	Shaking treatment	HO_SPC	3	11	72	0.153
## 28	Shaking treatment	HO_SPC	4	22	72	0.306
## 29	Shaking treatment	HO_SPC	5	40	72	0.556
## 30	Shaking treatment	HO_SPC	6	43	72	0.597
## 31	Shaking treatment	HO_MT	1	1	72	0.0139
## 32	Shaking treatment	HO_MT	2	4	72	0.0556
## 33	Shaking treatment	HO_MT	3	7	72	0.0972
## 34	Shaking treatment	HO_MT	4	20	72	0.278
## 35	Shaking treatment	HO_MT	5	29	72	0.403
## 36	Shaking treatment	HO_MT	6	34	72	0.472

10 Plots of cumulative incidence of tumoral development

10.1 Common themes

Theme of y-axis, and its presence or absence on the plots

```
scale_y <- scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.25),
                              labels = percent, position = "right")
```

Fix to hide y-axis numbers, ticks, and titles on the right and left

```
hide_y <- theme(axis.text.y.right = element_blank(),
                 axis.ticks.y.right = element_blank(),
                 axis.title.y.right = element_blank(),
                 axis.text.y = element_blank(),
                 axis.ticks.y = element_blank(),
                 axis.title.y = element_blank())
```

Fix to show y-axis numbers and ticks on the right

```
show_y <- theme(axis.text.y.right = element_text(color = "black", size = 14),
                 axis.ticks.y.right = element_line(),
                 axis.text.y = element_blank(),
                 axis.ticks.y = element_blank(),
                 axis.title.y = element_blank())
```

Theme of x-axis

```
scale_x <- scale_x_continuous(limits = c(0, 6), breaks = seq(0, 6, by = 2))
```

Color scales for treatment condition

```
treatment_scale <- list(
  scale_color_manual(values = treatment_condition_colors, breaks = c("Control", "Shaking
treatment")),
  scale_fill_manual(values = treatment_condition_colors, breaks = c("Control", "Shaking
treatment")))
```

Theme shared by all panels (angle = 90 for bottom-to-top reading)


```
theme_common <- theme(
  axis.text.x = element_text(color = "black", size = 14),
  axis.title.x = element_text(color = "black", size = 14, margin = margin(t = 10)),
  axis.title.y.right = element_text(color = "black", size = 14, angle = 90, margin =
margin(l = 15)),
  plot.margin = margin(l = 5, r = 5, t = 5, b = 5))
```

Specific label configurations for precise placement

```
labs_base <- labs(x = NULL, y = NULL, title = NULL)
labs_x_only <- labs(x = "Time (weeks)", y = NULL, title = NULL)
labs_y_only <- labs(x = NULL, y = "Cumulative incidence\nof tumor development", title =
NULL)
labs_xy <- labs(x = "Time (weeks)", y = "Cumulative incidence\nof tumor development",
title = NULL)
```

10.2 Panel A: by treatment condition

Cumulative incidence of tumor onset by treatment condition.

```
A <- cuminc(Surv(age_followup, factor(status_competing)) ~ treatment_condition,
  data = tum_df) %>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
  labs_xy + scale_y + treatment_scale + scale_x +
  theme_bw() + theme_common + show_y +
  theme(legend.position = "none") +
  plot_annotation(title = "Shaking treatment effect",
    theme = theme(plot.title = element_text(
      hjust = 0.5, face = "bold", size = 16, margin = margin(b = 10))))
```

Extract legend from plot A to use later on the full figure.

```
shared_legend <- cowplot::get_legend(
  A + theme(legend.position = "bottom",
    legend.direction = "horizontal",
    legend.text = element_text(size = 13),
    legend.title = element_text(size = 14, face = "bold")))
```

10.3 Panel B: by treatment condition for each lineage

10.3.1 HO_MT

Cumulative incidence of tumor onset by treatment condition for the lineage HO_MT.

```
B1 <- cuminc(Surv(age_followup, factor(status_competing)) ~ treatment_condition,
  data = tum_df %>% filter(lineage == "HO_MT")) %>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
```

```
labs_base + scale_y + scale_x + treatment_scale +
theme_bw() + theme_common + hide_y +
theme(legend.position = "none") +
annotate("text", x = 0.2, y = 0.95, label = expression(bolditalic("Ho") * bold("_MT")),
           color = "black", size = 5, hjust = 0, vjust = 1)
```

10.3.2 HO_SPC

Same method as above, but for the lineage HO_SPC.

```
B2 <- cuminc(Surv(age_followup, factor(status_competing)) ~ treatment_condition,
             data = tum_df %>% filter(lineage == "HO_SPC")) %>%
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_x_only + scale_y + scale_x + treatment_scale +
theme_bw() + theme_common + hide_y +
theme(legend.position = "none") +
annotate("text", x = 0.2, y = 0.95, label = expression(bolditalic("Ho") *
bold("_SPC")),
           color = "black", size = 5, hjust = 0, vjust = 1)
```

10.3.3 HO_SPT

Same method as above, but for the lineage HO_SPT.

```
B3 <- cuminc(Surv(age_followup, factor(status_competing)) ~ treatment_condition,
             data = tum_df %>% filter(lineage == "HO_SPT")) %>%
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_y_only + scale_y + scale_x + treatment_scale +
theme_bw() + theme_common + show_y +
theme(legend.position = "none") +
annotate("text", x = 0.2, y = 0.95, label = expression(bolditalic("Ho") *
bold("_SPT")),
           color = "black", size = 5, hjust = 0, vjust = 1)
```

Create panel B combining plots B1, B2, and B3

```
B <- (B1 + B2 + B3 + plot_layout(nrow = 1)) +
plot_annotation(title = "Interaction between the shaking treatment and the lineage",
               theme = theme(plot.title = element_text(hjust = 0.5, face = "bold",
                                                         size = 16, margin = margin(b =
                                                         10))))
```

10.4 Panel C: by lineage

10.4.1 HO_MT

Cumulative incidence of tumor onset for the lineage HO_MT.

```
C1 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
             data = tum_df %>% filter(lineage == "HO_MT")) %>%
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_base + scale_y + scale_x +
theme_bw() + theme_common + hide_y +
annotate("text", x = 0.2, y = 0.95, label = expression(bolditalic("Ho") * bold("_MT")),
           color = "black", size = 5, hjust = 0, vjust = 1)
```

10.4.2 HO_SPC

Same method as above, but for the lineage HO_SPC.

```
C2 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
             data = tum_df %>% filter(lineage == "HO_SPC")) %>%
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_x_only + scale_y + scale_x +
theme_bw() + theme_common + hide_y +
annotate("text", x = 0.2, y = 0.95, label = expression(bolditalic("Ho") *
bold("_SPC")),
           color = "black", size = 5, hjust = 0, vjust = 1)
```

10.4.3 HO_SPT

Same method as above, but for the lineage HO_SPT.

```
C3 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
             data = tum_df %>% filter(lineage == "HO_SPT")) %>%
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_y_only + scale_y + scale_x +
theme_bw() + theme_common + show_y +
annotate("text", x = 0.2, y = 0.95, label = expression(bolditalic("Ho") *
bold("_SPT")),
           color = "black", size = 5, hjust = 0, vjust = 1)
```

Create panel C combining plots C1, C2, and C3

```
C <- (C1 + C2 + C3 + plot_layout(nrow = 1)) +
plot_annotation(title = "Lineage effect",
               theme = theme(plot.title = element_text(
                 hjust = 0.5, face = "bold", size = 16, margin = margin(b = 10))))
```

10.5 Assembly

Add a letter tag (A/B/C) to the first sub-plot of each panel.

```

A <- A + labs(tag = "A") + theme(plot.tag = element_text(size = 20, face = "bold"))
B[[1]] <- B[[1]] + labs(tag = "B") + theme(plot.tag = element_text(size = 20, face =
"bold"))
C[[1]] <- C[[1]] + labs(tag = "C") + theme(plot.tag = element_text(size = 20, face =
"bold"))

```

Convert each panel into a graphical object. (Wrapping A in patchwork to force perfect alignment with the B and C grids).

```

A <- patchwork::patchworkGrob(A + patchwork::plot_layout(ncol = 1))
B <- patchwork::patchworkGrob(B)
C <- patchwork::patchworkGrob(C)

```

Combine the panels and the shared legend vertically. The legend is placed between B and C.

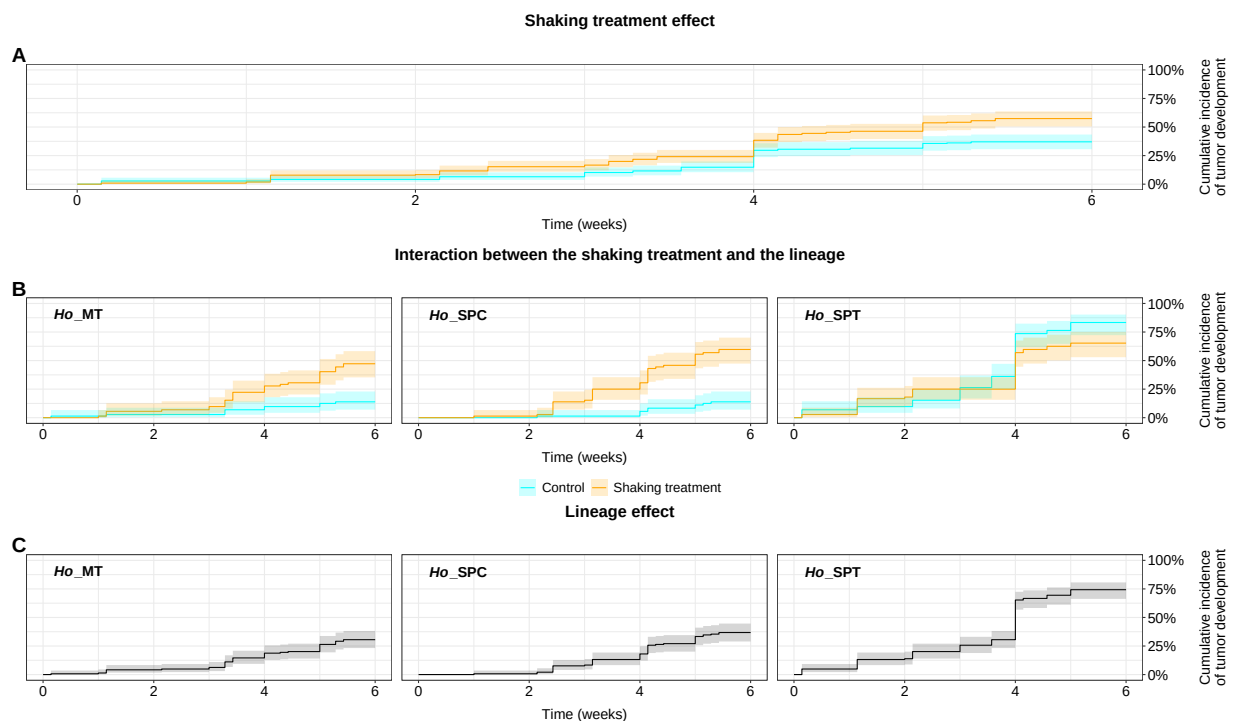
```

final_tum_panel <- cowplot::plot_grid(A,
                                     B,
                                     shared_legend,
                                     C,
                                     ncol = 1,
                                     align = "v",
                                     axis = "lr",
                                     rel_heights = c(1, 1, 0.1, 1)) +
  theme(plot.margin = margin(l = 5, r = 5, t = 5, b = 5))

```

Show the final figure.

```
print(final_tum_panel)
```



```
ggsave("figure_2.png", plot = final_tum_panel, width = 21, height = 29.7, units = "cm",  
dpi = 300)
```